

Replication and Barren Plateau Analysis of Quantum Long Short-Term Memory Networks

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1. Introduction

Quantum Machine Learning (QML) has emerged as a compelling intersection of quantum computing and classical machine learning, leveraging superposition and entanglement for pattern modelling. Among the proposed hybrid architectures, the **Quantum Long Short-Term Memory (QLSTM)** network [1] replaces parametric linear transformations inside classical LSTM cells with *Variational Quantum Circuits* (VQCs), offering theoretical expressibility advantages for sequence modelling.

However, a fundamental and underexplored challenge in all VQC-based models is the **Barren Plateau (BP) phenomenon**. The seminal survey by Larocca et al. [2] establishes that *every component* of a variational quantum algorithm—the ansatz, initial state, observable, loss function, and hardware noise—can independently cause BPs, where the optimisation landscape becomes exponentially flat and featureless as system size grows. This is not merely a theoretical concern: on a BP, an exponentially large number of measurement shots is required to identify any loss-minimising direction, making training infeasible in practice. Despite this, the trainability of QLSTM under realistic circuit configurations has not been rigorously characterised.

This project closes that gap. I perform a faithful replication of the QLSTM architecture and then conduct a systematic empirical study of BP onset within its VQC sub-circuits. Crucially, I evaluate and compare three concrete mitigation strategies, including the recent two-stage convex-initialisation framework of Boabang & Gyamerah [5], which shapes the quantum energy landscape into a smooth basin before nonconvex refinement.

The practical motivation is concrete: before investing resources in QLSTM for real-world tasks like **stock price forecasting** and **weather prediction**, a practitioner needs an evidence-based answer to—*does this model actually train, and under what circuit configurations?*

2. Project Description

The project is structured in two tightly coupled phases.

Phase 1 — Faithful Replication

I will implement the QLSTM architecture from Chen et al. [1] using PennyLane and PyTorch. Each LSTM gate’s linear transformation is replaced by a VQC sub-circuit using an angle-embedding layer followed by a strongly entangling ansatz. The primary benchmark is **stock price prediction** using daily closing prices from Yahoo Finance (e.g., S&P 500 index). Weather forecasting (Delhi or Jena Climate dataset) is included as a stretch goal if time per-

mits. A classical LSTM baseline with a *matched parameter count* is trained for fair comparison on the primary benchmark.

Phase 2 — Barren Plateau Analysis & Mitigation

Guided by the taxonomy in Larocca et al. [2], I identify the specific BP sources active in QLSTM (ansatz expressibility and random initialisation) and measure **gradient variance** across VQC sub-circuits as a function of:

- Circuit depth: $L \in \{1, 2, 4\}$
- Qubit count: $n \in \{2, 4, 6\}$

Gradient variance is estimated over 200 random parameter initialisations per configuration. I will then implement and compare **two** mitigation strategies:

1. **Identity-block initialisation** (Grant et al. [4]) — sub-circuits initialised so the net unitary is close to identity, keeping initial gradients non-negligible.
2. **Two-stage least squares** (Boabang & Gyamerah [5]) — a convex initialisation stage shapes the loss landscape into a smooth low-energy basin, followed by non-convex refinement. This is the most recent (2026) mitigation framework in the literature.

AI coding agents (as permitted by the course) will be used to accelerate boilerplate circuit construction and experiment scripting.

Papers to Implement / Survey

1. Chen et al. (2020). *Quantum Long Short-Term Memory*. [arXiv:2009.01783](https://arxiv.org/abs/2009.01783) — primary architecture to replicate.
2. Larocca et al. (2024). *Barren Plateaus in Variational Quantum Computing: A Review*. [arXiv:2405.00781](https://arxiv.org/abs/2405.00781) — unified theoretical framework for BP sources and mitigations; primary reference for Phase 2 design.
3. Boabang & Gyamerah (2026). *Overcoming Barren Plateaus via a Two-Step Least Squares Approach*. [arXiv:2601.18060](https://arxiv.org/abs/2601.18060) — mitigation strategy 2; novel two-stage optimisation framework.

3. Expected Outcome

By the end of the project, I will produce:

- A working PennyLane/PyTorch QLSTM implementation reproducing the key results of Chen et al. on the stock price benchmark.
- Gradient variance plots across circuit depths $L \in \{1, 2, 4\}$ and qubit counts $n \in \{2, 4, 6\}$, characterising BP onset and cross-referenced with the taxonomy of Larocca et al. [2].

- A comparison of **two** mitigation strategies (identity-block initialisation and two-stage least squares) with training loss curves and final MAE/RMSE.
- A quantitative QLSTM vs. classical LSTM comparison table: prediction error, parameter count, and training time.
- A clear, honest assessment of which circuit configurations make QLSTM trainable, grounded in BP theory.

4. Midterm Deliverables

- Reading notes for all three papers, with a written summary of the BP taxonomy from Larocca et al.
- Working QLSTM implementation that trains on a toy dataset (sine wave) confirms circuit wiring and gradient flow are correct.
- Classical LSTM baseline with matched parameter count on the stock price dataset.
- Preliminary gradient variance measurements at $n \in \{2, 4\}$ qubits and $L \in \{1, 2\}$ layers, confirming the experimental setup is reproducible.

5. Final Deliverables

- Full QLSTM replication on the stock price dataset with a classical LSTM comparison table (MAE, RMSE, parameter count, training time). Weather forecasting included as a stretch goal.
- Complete BP analysis: gradient variance plots across $L \in \{1, 2, 4\}$ and $n \in \{2, 4, 6\}$, annotated using BP-source taxonomy from Larocca et al.
- Comparison of both mitigation strategies (identity-block initialisation and two-stage least squares) with training loss curves and final accuracy numbers.
- Documented code repository with reproducible scripts, ready for demo during viva.
- Final report and presentation summarising all results and conclusions.

Progress Report

6. Papers Read

1. Quantum Long Short-Term Memory

Authors: Samuel Yen-Chi Chen, Shinjae Yoo, Yao-Lung L. Fang

Venue: arXiv preprint, arXiv:2009.01783 (2020)

The paper proposes QLSTM by replacing each linear transformation in the four LSTM gates (forget, input, cell, output) with a Variational Quantum Circuit (VQC). Each VQC first prepares qubits from the ground state using angle embedding (analogous to QFT/Hadamard for state preparation), then applies parameterised strongly entangling layers. Gradient computation uses the parameter shift rule ($\pm\pi/2$ shifts), which integrates with classical backpropagation via the chain rule to optimise the full hybrid model end-to-end.

2. Barren Plateaus in Variational Quantum Computing: A Review

Authors: Martin Larocca, Supanut Thanasilp, Samson Wang, Kunal Sharma, Jacob Biamonte, Patrick J. Coles, Lukasz Cincio, Jarrod R. McClean, Zoë Holmes, M. Cerezo

Venue: arXiv preprint, arXiv:2405.00781 (2024)

This comprehensive review establishes a unified taxonomy of barren plateau (BP) sources in variational quantum algorithms. The core result is that the variance of the loss gradient $\text{Var}[\partial L / \partial \theta]$ shrinks exponentially as $O(1/2^n)$ with qubit count n , making gradient-based optimisation infeasible for large circuits. BPs arise independently from ansatz expressibility, entanglement structure, noise, and choice of global vs. local cost functions.

3. Overcoming Barren Plateaus via a Two-Step Least Squares Approach

Authors: Francis Boabang, Samuel A. Gyamerah

Venue: arXiv preprint, arXiv:2601.18060 (2026)

Proposes a two-stage optimisation framework to escape barren plateaus. Stage 1 uses regularised least squares (a convex problem) to find a good initial parameter region, shaping the loss landscape into a smooth basin. Stage 2 refines these parameters using unregularised gradient-based optimisation. This avoids the problem of random initialisation landing on exponentially flat regions of the landscape.

4. Barren Plateaus in Quantum Neural Network Training Landscapes

Authors: Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, Hartmut Neven

Venue: *Nature Communications*, 9(1):4812 (2018)

Seminal paper providing the first rigorous proof that for sufficiently random parameterised quantum circuits, gradient magnitudes vanish exponentially with qubit count due to concentration of measure. Establishes the mathematical foundation showing why deep, randomly initialised VQCs become untrainable, the result that motivates all subsequent BP mitigation research.

5. An Initialization Strategy for Addressing Barren Plateaus

Authors: Edward Grant, Leonard Wossnig, Mateusz Ostaszewski, Marcello Benedetti

Venue: *Quantum*, 3:214 (2019)

Proposes initialising VQC parameters such that each circuit block approximates the identity operation, analogous to classical He/Xavier initialisation. This ensures the circuit starts in a non-flat region of the optimisation landscape with non-vanishing gradients, enabling successful training before the circuit explores more expressive regions.

7. Summary of Key Ideas

The papers collectively address the challenge of training variational quantum circuits for machine learning. Three key themes emerge:

1. **Hybrid Quantum-Classical Architectures:** Chen et al. demonstrate that VQCs can replace linear layers inside classical neural networks (specifically LSTMs). The parameter shift rule enables gradient computation through quantum circuits, which integrates with classical backpropagation via the chain rule—the $\pm\pi/2$ shift computes $\partial\langle O \rangle / \partial\theta$ (the quantum part), and PyTorch autograd handles $\partial L / \partial\langle O \rangle$ (the classical loss part).
2. **The Barren Plateau Problem:** McClean et al. (2018) and Larocca et al. (2024) establish that randomly initialised VQCs suffer from exponentially vanishing gradients as circuit size grows. This is a fundamental obstacle: simply scaling up quantum circuits does not improve performance without careful design. BP sources include circuit expressibility (too expressive $\Rightarrow$ approximate random unitary $\Rightarrow$ flat landscape), entanglement, hardware noise, and global cost functions.
3. **Mitigation Through Initialisation:** Both Grant et al. and Boabang & Gyamerah address BPs through smarter initialisation. Grant et al. use identity-block initialisation (keeping initial unitaries near identity, analogous to classical He/Xavier), while Boabang & Gyamerah use a two-stage approach where convex optimisation first finds a good starting region. Both ensure the optimiser begins where gradients are meaningful.

8. Implementation Status

The following components have been implemented using PennyLane (v0.42) and PyTorch (v2.6):

1. **QLSTM Architecture** (`src/qlstm.py`) — VQC sub-circuits with angle embedding + strongly entangling layers, QLSTM cell with four VQC-based gates, full model wrapper.
2. **Classical LSTM Baseline** (`src/classical_lstm.py`) — standard PyTorch LSTM for fair comparison.
3. **Sine Wave Training** (`src/train_sine.py`) — toy benchmark that confirms correct circuit wiring and gradient flow.
4. **S&P 500 Stock Training** (`src/train_stock.py`) — daily closing-price prediction using `yfinance` data, training all three QLSTM variants and the classical baseline.
5. **Barren Plateau Analysis** (`src/bp_full_grid.py`) — gradient variance over 200 random initialisations for the full proposal grid $n \in \{2, 4, 6\}$, $L \in \{1, 2, 4\}$.
6. **Mitigation Strategies** (`src/mitigations.py`) — identity-block initialisation (Grant et al.) and two-stage LS warm-up (Boabang & Gyamerah).
7. **Trainability Table** (`src/trainability_table.py`) — automatic classifier mapping each (n, L) to a train-

ability verdict.

Repository: <https://github.com/gautam2905/IQC-Project>

Barren Plateau Characterisation

Using 200 random initialisations per configuration and taking the average variance across all circuit parameters, we obtained the following for the VQC sub-circuit used inside QLSTM gates:

Layers \ Qubits	$n=2$	$n=4$	$n=6$
$L=1$	8.82×10^{-2}	2.61×10^{-2}	5.58×10^{-3}
$L=2$	6.57×10^{-2}	1.40×10^{-2}	8.53×10^{-3}
$L=4$	8.00×10^{-2}	2.35×10^{-2}	4.70×10^{-3}

Going from $n=2$ to $n=6$ qubits, the gradient variance shrinks by $\sim 15\times$ — consistent with the exponential $O(1/2^n)$ decay predicted by McClean et al. [3] and Larocca et al. [2].

Trainability Verdict per Configuration

Using thresholds (Trainable $\geq 10^{-2}$, Marginal 10^{-4} – 10^{-2} , Barren $< 10^{-4}$):

Layers \ Qubits	$n=2$	$n=4$	$n=6$
$L=1$	Trainable	Trainable	Marginal
$L=2$	Trainable	Trainable	Marginal
$L=4$	Trainable	Trainable	Marginal

At $n=6$, gradient magnitudes are already in the “marginal” regime, foreshadowing a barren plateau for slightly larger circuits. Practitioners should target $n \leq 4$ for reliable training without mitigation.

S&P 500 Stock Price Results

Training configuration: sequence length 10, 80 training samples, 20 epochs, Adam (lr = 0.02). QLSTM: $n_q=2$, $L=1$, hidden = 4.

Model	MAE	RMSE	Params	Time
Classical LSTM (h=8)	2.363	2.367	361	2m
Classical LSTM matched (h=4)	2.000	2.001	117	2m
QLSTM (random init)	1.242	1.249	125	5s
QLSTM (identity-block init)	1.246	1.253	125	5s
QLSTM (two-stage LS)	2.374	2.389	125	5s

The parameter-matched comparison uses Classical LSTM with `hidden_size=4` (117 params) against QLSTM with $n_q=2$, $L=1$ (125 params) — a near-identical parameter budget. Three observations:

- **QLSTM beats the size-matched classical baseline by $\sim 37\%$ in MAE** (1.242 vs 2.000) with essentially the same parameter count (125 vs 117). This is a strong result: at small circuit sizes, QLSTM’s

quantum expressibility provides a genuine advantage over an equivalently sized classical model on this non-stationary time series.

- **Identity-block initialisation behaves nearly identically to random init** at $n=2$. This is expected: barren plateaus are not yet severe at two qubits ($\text{Var} \sim 10^{-1}$), so smart initialisation provides little benefit in this regime. Its value should become apparent at larger n where BP is a real obstacle.
- **Two-stage LS warm-up degrades generalisation.** Stage 1 finds an analytically optimal linear projection onto the current (random) quantum features; however, subsequent gradient descent perturbs those quantum features, and the combination overfits the training window of the non-stationary S&P 500 series. This is an *honest null result*: the convex warm-up may help *optimisation speed* but can hurt *generalisation* under distribution shift. A regularised variant of the warm-up is proposed as the novel extension.

Sine Wave Sanity Check

The earlier toy benchmark (sine wave, 50 train samples, 30 epochs) confirmed correct QLSTM gradient flow: loss decreased monotonically from 0.52 to 0.48; classical LSTM reached 0.24 on the same task with $\sim 4\times$ more parameters.

9. Refined Final Deliverable

All core deliverables from the original proposal are now implemented; the remaining work is empirical scaling and a novel extension.

Completed

1. QLSTM replication with angle embedding + strongly entangling ansatz.
2. Classical LSTM baseline with matched architecture.
3. BP gradient-variance sweep across the proposal grid $n \in \{2, 4, 6\}$, $L \in \{1, 2, 4\}$.
4. Identity-block initialisation and two-stage LS warm-up mitigations.
5. Full S&P 500 benchmark comparing Classical LSTM against three QLSTM variants, with MAE / RMSE / parameter count / training-time table.
6. Trainability verdict table mapping each (n, L) pair to Trainable / Marginal / Barren.

Planned (remaining)

1. **Scaling the BP sweep** from $n=6$ to $n \approx 20$ qubits, producing a clean log-linear $O(1/2^n)$ curve. This is purely a compute-time investment on the classical simulator.
2. **Extended stock training**: larger training windows, more epochs, and error bars across multiple seeds for the Classical vs. QLSTM comparison. The current small-window run (80 samples, 20 epochs) was chosen

to keep quantum-simulation cost manageable during the midterm phase.

3. **Weather forecasting stretch goal** (Delhi / Jena Climate) as a secondary benchmark if time permits.
4. **Novel extension (hypothesis, scope-dependent)**: our S&P 500 experiments suggest that two-stage LS warm-up can *hurt* generalisation under distribution shift — it fits the training window tightly but becomes brittle downstream. We plan to investigate a *regularised variant* of the warm-up that keeps the good training-time properties (smooth basin for Stage 2) while explicitly preserving generalisation, e.g. by combining the LS warm-up with an early-stopping criterion or a regularisation term tied to the Fisher information of the classical readout. A positive result would constitute a small but concrete contribution beyond pure replication.
5. **Viva-ready notebook** bundling the full pipeline for reproducible demo.

References

- [1] S. Chen, S. Yoo, and Y.-L. Fang. Quantum long short-term memory. *arXiv preprint arXiv:2009.01783*, 2020. <https://arxiv.org/abs/2009.01783>
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